

Part 2 Examples of Transportation Problems

Transportation Problem – Part 2: Examples

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Part 2: Examples of Transportation Problems

Educational Series on Transportation Problems *Understanding Through Practical Examples*

This document provides five detailed examples of transportation problems, ranging from balanced to unbalanced scenarios, with step-by-step results using North-West Corner (NWC), Least Cost Method (LCM), Vogel's Approximation Method (VAM), and Modified Distribution (MODI) methods.

1. Introduction: Why Examples Are Important for Understanding TP

Transportation problems (TP) are a fundamental class of linear programming problems that deal with the optimal distribution of goods from multiple sources to multiple destinations. While the theory behind TP is essential, practical examples bring the concepts to life. Examples help learners:

- Understand the structure of supply, demand, and cost matrices.
- Differentiate between balanced and unbalanced problems.
- Apply various solution methods (NWC, LCM, VAM, MODI) in real contexts.
- Compare results and see how different methods affect the total cost.
- Recognize when dummy sources or destinations are needed.

Each example in this document is fully worked out, with clear cost tables and final results for each method.

2. Example 1: Balanced 3×4 Problem

Problem Description

A balanced transportation problem has total supply equal to total demand. Here, we have 3 sources and 4 destinations.

Supply: Source 1 = 10, Source 2 = 15, Source 3 = 20 Demand: Destination A = 8, Destination B = 12, Destination C = 10, Destination D = 15 Total Supply = 45, Total Demand = 45 Balanced

Cost Matrix (Cost per unit)

Source \ Destination

A

B

C

D

Supply

Source 1

5

3

8

6

10

Source 2

4

7

2

5

15

Source 3

6

4

9

3

20

Demand

8

12

10

15

45

Results

Method

Total Cost

North-West Corner (NWC)

266

Least Cost Method (LCM)

135

Vogel's Approximation (VAM)

132

Modified Distribution (MODI)

132

Explanation: A balanced problem means the sum of all supplies equals the sum of all demands. No dummy row or column is needed. The MODI result of 132 is the optimal solution.

3. Example 2: Unbalanced (Excess Supply) 3×4 Problem

Problem Description

When total supply exceeds total demand, a dummy destination is added with zero cost to absorb the excess supply.

Supply: Source 1 = 30, Source 2 = 25, Source 3 = 20 Demand: Destination A = 20,
Destination B = 15, Destination C = 10, Destination D = 12 Total Supply = 75, Total
Demand = 57 Excess Supply = 18

Cost Matrix (with Dummy Destination)

Source \ Destination

A

B

C

D

Dummy

Supply

Source 1

8

6

9

7

0

30

Source 2

5

4

8

6

0

25

Source 3

7

9

3

5

0

20

Demand

20

15

10

12

18

75

Results

Method

Total Cost

North-West Corner (NWC)

488

Least Cost Method (LCM)

333

Vogel's Approximation (VAM)

320

Modified Distribution (MODI)

320

Explanation: The dummy destination has zero transportation cost. It represents the unused supply that is not shipped. The MODI result of 320 is optimal.

4. Example 3: Unbalanced (Excess Demand) 2×4 Problem

Problem Description

When total demand exceeds total supply, a dummy source is added with zero cost to meet the excess demand.

Supply: Source 1 = 8, Source 2 = 10 Demand: Destination A = 6, Destination B = 7, Destination C = 5, Destination D = 4 Total Supply = 18, Total Demand = 22 Excess Demand = 4

Cost Matrix (with Dummy Source)

Source \ Destination

A

B

C

D

Supply

Source 1

3

5

7

4

8

Source 2

6

2

8

5

10

Dummy Source

0

0

0

0

4

Demand

6

7

5

4

22

Results

Method

Total Cost

North-West Corner (NWC)

117

Least Cost Method (LCM)

83

Vogel's Approximation (VAM)

79

Modified Distribution (MODI)

79

Explanation: The dummy source has zero production cost. It represents unmet demand that is not supplied. The MODI result of 79 is optimal.

5. Example 4: Small 2×3 Problem

Problem Description

A small balanced problem with 2 sources and 3 destinations.

Supply: Source 1 = 12, Source 2 = 10 Demand: Destination A = 8, Destination B = 7,

Destination C = 7 Total Supply = 22, Total Demand = 22 Balanced

Cost Matrix

Source \ Destination

A

B

C

Supply

Source 1

4

6

9

12

Source 2

7

3

5

10

Demand

8

7

7

22

Results

Method

Total Cost

North-West Corner (NWC)

126

Least Cost Method (LCM)

109

Vogel's Approximation (VAM)

109

Modified Distribution (MODI)

109

Explanation: This small example shows that LCM and VAM can sometimes yield the same result, which is also optimal (MODI).

6. Example 5: Large 4×5 Problem

Problem Description

A larger balanced problem with 4 sources and 5 destinations.

Supply: Source 1 = 20, Source 2 = 25, Source 3 = 30, Source 4 = 15 Demand: Destination A = 18, Destination B = 22, Destination C = 12, Destination D = 20, Destination E = 18 Total Supply = 90, Total Demand = 90 Balanced

Cost Matrix

Source \ Destination

A

B

C

D

E

Supply

Source 1

7

5

9

6

8

20

Source 2

4

8

3

7

5

25

Source 3

6

4

8

5

9

30

Source 4

5

7

6

4

3

15

Demand

18

22

12

20
18
90

Results

Method

Total Cost

North-West Corner (NWC)

722

Least Cost Method (LCM)

448

Vogel's Approximation (VAM)

434

Modified Distribution (MODI)

434

Explanation: This larger example demonstrates that VAM often provides a solution very close to or equal to the optimal MODI solution.

7. Summary Comparison Table

Example

Type

Size

NWC

LCM

VAM

MODI (Optimal)

1

Balanced

3×4

266

135

132

132

2

Unbalanced (Excess Supply)

3×4

488

333

320

320

3

Unbalanced (Excess Demand)

2×4

117

83

79

79

4

Small Balanced

2×3

126

109

109

109

5

Large Balanced

4×5

722

448

434

434

Observations:

- NWC consistently gives the highest cost because it ignores cost values.
- LCM improves significantly over NWC.

- VAM often matches the optimal MODI solution, especially in larger problems.
 - MODI confirms the optimal solution in all cases.
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8. Key Takeaways

1. Balanced vs. Unbalanced: Always check if total supply equals total demand. If not, add a dummy row (excess demand) or dummy column (excess supply) with zero cost.
 2. Method Selection:
 - NWC is simple but inefficient.
 - LCM is better but may not be optimal.
 - VAM is highly effective and often optimal.
 - MODI is the definitive method to find the optimal solution.
 3. Dummy Variables: Dummy sources or destinations have zero cost and represent unmet demand or unused supply.
 4. Practical Use: Transportation problems are widely used in logistics, supply chain management, and resource allocation.
 5. Optimality: The MODI method guarantees the minimum total transportation cost, making it the gold standard for solving TP.
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End of Part 2

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